



Distributed Information Processing

3rd Lecture

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Outline

- Clock and Global States
 - Global States
 - Determining Consistent Global States
- Q&A

Global States

- Prefix of P_i 's History & Global History

$$h_i^k = \langle e_i^j \mid j = 1, \dots, k \rangle, \quad H = \bigcup_{i=1}^N h_i$$

- Cut & Frontier

$$C = \bigcup_{i=1}^N h_i^{C_i}, \quad F = \{e_i^{C_i} \mid i = 1, \dots, N\}$$

Set of All
Affected Values

- Global State (Corresponding to C)

$$S = \{s_i^{C_i} \mid i = 1, \dots, N, s_i^{C_i} \text{ is } P_i\text{'s state immediately after } e_i^{C_i}\}$$

- Run: a Total Ordering in a Global History
Consistent with Each Local History

Consistent Cuts & Runs

- C Is Consistent If the Following Holds:

$$\forall e \in C, e' \rightarrow e \Rightarrow e' \in C$$

H Is Consistent If the
Corresponding C Is Consistent

Figure 14.9 on Page 629

Coulouris, Dollimore and Kindberg Distributed Systems: Concepts and Design Edn. 5 © Pearson Education 2012

- Consistent Run: a Total Ordering in a Consistent Global History, Consistent with the Happened-Before Relation

Lattice of Global States

■ Observing Consistent Global States

$S = \{s_i \mid i = 1, \dots, N\}$ Is Consistent iff $VC_i(s_i)[i] \geq VC(s_j)[i]$ for $i, j = 1, \dots, N$

S_{ij} Is in
Level $(i+j)$

S_{ij} = global state after i events at process 1
and j events at process 2

$(1,0)$ $(2,0)$

Figure 14.14 on Page 637

Figure 14.15 on Page 638

Physical
time

$(2,1)$

Inconsistent Cut,
Inconsistent S_{01}

Inconsistent Cut,
Inconsistent S_{11}

Not
Reachable

S_{30} S_{22}



Detecting Global Properties

Figure 14.8 on Page 627

Distributed 'Snapshot' Algorithm [Chandy85]

■ Consistent Global-State Detection

Marker to Record the State

Figure 14.10 on Page 632

Assumptions

- Reliable, Strongly-Connected Components
- Unidirectional Channels & In-Order Message Delivery

Illustration: How the Alg. Works

- Initial States of the Components



Figure 14.11 on Page 632

P2 Has Already Received an Order of Five Widgets

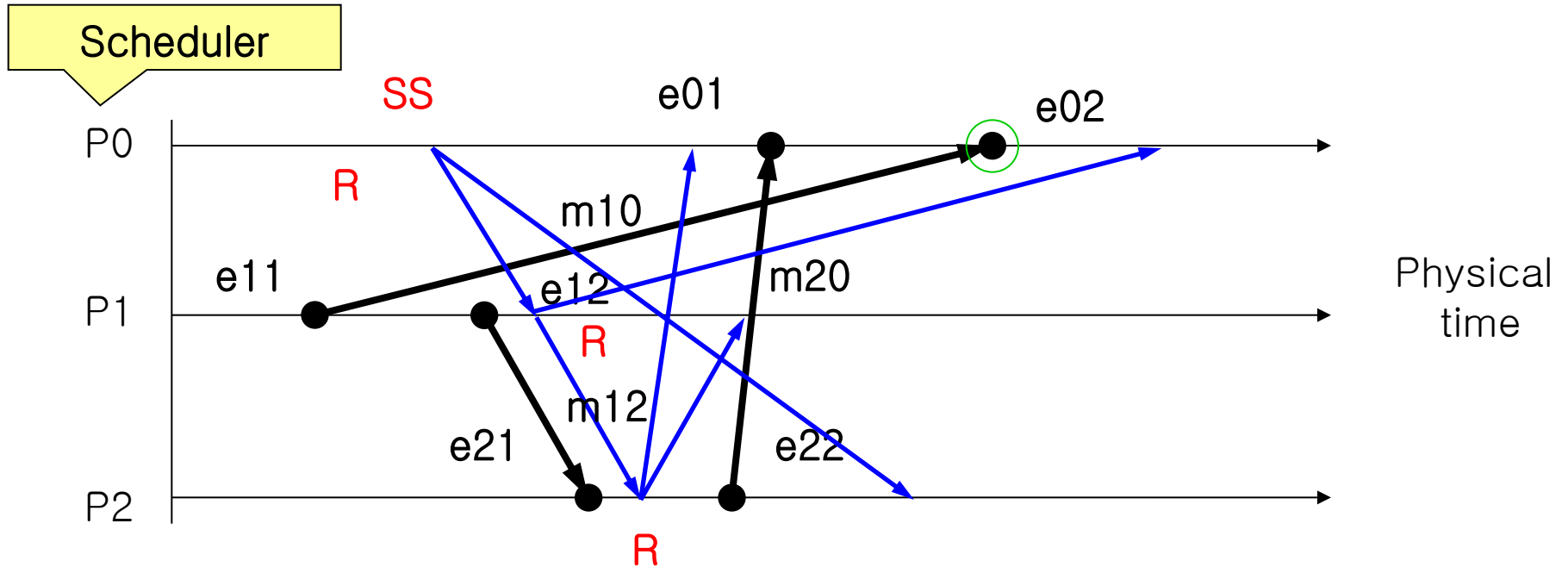
Illustration (Cont'd)

P2 Has Already
Received an Order
of Five Widgets

Figure 14.12 on Page 633

■ $S = p1: \langle \$1000, 0 \rangle; p2: \langle \$50, 1995 \rangle;$
 $c1: \langle (\text{five widgets}) \rangle; c2: \langle \rangle$

Illustration w/ a Diagram

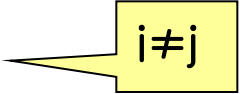


- SS: $p0: \langle \rangle$; $p1: \langle e11, e12 \rangle$; $p2: \langle e21 \rangle$
 $c01: \langle \rangle$; $c02: \langle \rangle$; $c10: \langle m10 \rangle$; $c12: \langle \rangle$
 $c20: \langle \rangle$; $c21: \langle \rangle$

Consistency Proof

- States Recorded by the Alg. Are Consistent:

$$\forall e_j \in C, e_i \rightarrow e_j \Rightarrow e_i \in C$$

Show: $e_i \notin C, e_i \rightarrow e_j \Rightarrow e_j \notin C$ 

- Assume That P_i Recorded Its State before e_i
- Marker Would Have Reached P_j before the Message for e_j
- P_j Would Have Recorded Its State before e_j

Characteristics of Snapshots

■ Derivation of “Observed” Run from “Actual” Run

"Actual" Run : $\langle e_i^k \mid i = 1, \dots, N \rangle = \langle e^j \rangle$

"Permuted" Run : $\langle \dots, e^{R-1}, e^R, \dots \rangle$

Recoding

Observed
(Consistent) Run

- A Non-Observed Event May Occur before an Observed Event in the “Actual” Run
- If a Non-Observed Event Precedes an Observed Event (Next to it) in the “Actual” Run, the Events Can Be Swapped Preserving Consistency

Global State Predicates

- Functions That Map Global States to True or False
 - Stable: Once True, It Remains True
 - E.g., deadlock or termination
 - Unstable: Not Stable
 - Possibly True: True At Some Point
 - E.g., snapshot by the 'Snapshot' Algorithm
 - Definitely True: True in All Cases